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Bihar Engineering University, Patna

B.Tech 2nd Semester Special Exam., 2020

Time: 3 hours

Code: 100103/100203 ***Subject: Chemistry*** ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. I is compulsory.*

Q.1 Answer any seven questions in brief: **[2x7=14]**

(a) Arrange the following in increasing order of stability:
$N_2, N_2^+, N_2^-, N_2^{2-}$

(b) Transition metal ions like Cu^+ and Ag^+ are colourless. Why?

(c) Which of Cr^+ or Cu^+ is expected to be coloured?

(d) ^{13}C is NMR active, but ^{12}C is not. Why?

(e) What is the direction of a reaction when $\Delta G = 0$?

(f) Why is work not a state function?

(g) Write the relationship between parts per million [ppm] and Clarke's degree [°C].

(h) What is critical temperature of a gas?

(i) Arrange the following ligands in order of increasing field strength:
CN^-, CO, H_2O, NH_3

(j) Arrange the following in order of their increasing reactivity in nucleophilic substitution reaction:
$CH_3F, CH_3I, CH_3Br, CH_3Cl$

Q.2 Solve all parts : **[4+4+6=14]**

(a) At what temperature will water boil when the applied pressure is 528 mm of Hg? (Latent heat of vaporisation of water = 545.5 cal/g)

(b) At NTP, 2.8 L of O_2 were mixed with 19.6 L of H_2. Calculate the increase in entropy (assume ideal gas behaviour).

(c) The equilibrium constants for the reaction $H_2[g] + S(s) \rightarrow H_2S[g]$ are 18.5 at 925 K and 9.25 at 1000 K. Calculate standard enthalpy of the reaction. Also calculate $\Delta G\backslash circ$ and $\Delta S\backslash circ$ at 925 K.

Q.3 Solve all parts : **[4+5+5=14]**

(a) The uncertainties in the position and velocity of a particle are 9.5×10^{-10}m and $5.5\times10^{-20} ms^{-1}$, respectively. Calculate the mass of the particle. ($h = 6.626\times10^{-34}$J·s)

(b) Calculate the kinetic energy of a moving electron which has a wavelength of 48 pm. [Mass of electron = 9.11×10^{-31}kg]

(c) Discuss the failures of classical mechanics to explain properties of particles at atomic and sub-atomic levels.

Q.4 Solve both questions : **[8+6=14]**

(a) Draw the MO energy-level diagram for O_2 and based on the diagram, and explain the magnetic property observed in O_2, O_2^+ and O_2^-.

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(b) Explain geometrical isomerism and optical isomerism for transition metal complex with an example for each.

Q.5 Solve all parts : **[4+4+6=14]**

(a) The internuclear distance of NaCl is 2.36×10^{-10}m. Calculate the reduced mass and moment of inertia of NaCl. (Atomic mass of Cl = 35×10^{-3}kg mol^{-1} and Na = 23×10^{-3}kg mol^{-1})

(b) Calculate the force constant for CO, if it absorbs at 2.143×10^{5}m^{-1}. (Atomic mass of C = 12×10^{-3}kg mol^{-1} and O = 16×10^{-3}kg mol^{-1})

(c) How many ^{1}H NMR signals are there in—

(i) CH_3-CH_3
(ii) $CH_3-CH_2-CH_3$
(iii) CH_3-CH_2-Cl
(iv) $CH_3-CHCl-CH_3$
(v) $C_6H_5CH_3$
(vi) $C_6H_5CH_2CH_3$

Q.6 Solve both questions : **[5+9=14]**

(a) 2 mole of NH_3 at 300 K occupy a volume of 5×10^{-3}m^3. Calculate the pressure using van der Waals equation (a = 0.417N m^{4}mol^{-2} and $b = 0.037\times10^{-3}$m^{3}mol^{-1}). Compare the above result with the pressure calculated using ideal gas equation.

(b) Write short notes on the following:

(i) Magnetic resonance imaging
(ii) Fingerprint region in infrared spectroscopy
(iii) Different types of electronic excitations

Q.7 Solve both questions : **[7+7=14]**

(a) Consider the following half-cell reactions:

$PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e \rightarrow$
$PbSO_4(s) + 2H_2O$, $E\backslash circ = 1.70V$
$PbSO_4(s) + 2e \rightarrow Pb(s) + SO_4^{2-}(aq)$, $E\backslash circ = -0.31V$

Write the cell (in proper cell notation) and the cell reaction. Calculate the value of $E\backslash circ$ for the cell and the EMF generated if $[H^+] = 0.1M$ and $[SO_4^{2-}] = 2M$.

(b) A water sample had the following constituents per litre:

$CaCO_3$= 81mg, $MgHCO_3$= 75mg, $CaSO_4$= 136mg, $MgSO_4$= 120mg, $NaCl$= 47mg

Calculate the quantity of temporary and permanent hardness in the water sample. Calculate the quantity of lime (78% purity) and soda (92% purity) required for softening of 1.5 million litres of the above water sample.

Q.8 Solve all parts : **[4+5+5=14]**

(a) Describe two methods used for resolving racemic mixtures into optically active compounds.

(b) Write the possible optical isomers of tartaric acid and indicate the point of symmetry or plane of symmetry (if any) in the isomers.

(c) Differentiate between (i) enantiomers and diastereomers and (ii) racemic mixture and meso compounds.

Q.9 Solve all parts : **[4+4+6=14]**

(a) How do you decide whether the reaction $CH_3Br + OH^- \leftharpoons CH_3OH + Br^-$ proceeds by S_N1 or S_N2 reaction? Give justification in favour of your answer.

(b) Draw the energy profile diagram for the following reaction:

$(CH_3)_2CBr + OH^- \leftharpoons (CH_3)_2COH + Br^-$

(c) Write short notes on the following:

(i) Steric effects
(ii) Diels-Alder reaction

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